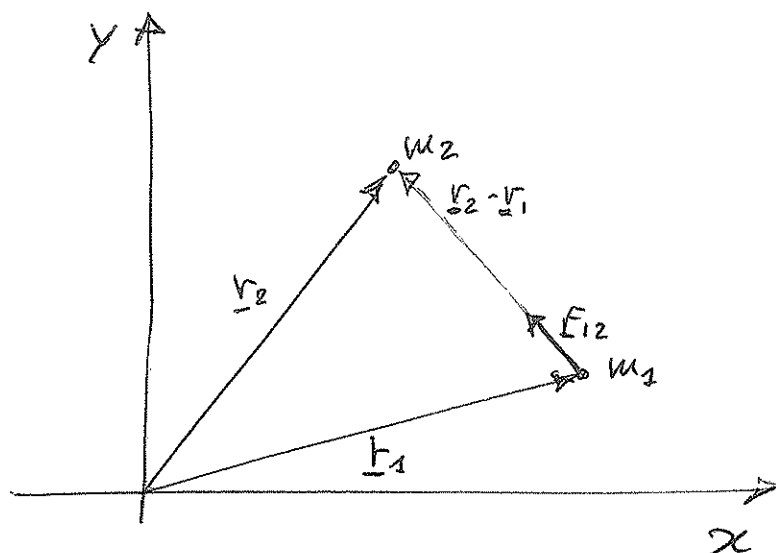


# THREE-BODY PROBLEM



$$\underline{r}_1 = x_1 \hat{i} + y_1 \hat{j}$$

$$\underline{r}_2 = x_2 \hat{i} + y_2 \hat{j}$$

$$\underline{r}_3 = x_3 \hat{i} + y_3 \hat{j}$$

La forza esercitata su  $m_2$  da  $m_1$  è:

$$F_{12} = m_1 m_2 \frac{(\underline{r}_2 - \underline{r}_1)}{|\underline{r}_2 - \underline{r}_1|^3}$$

In generale la forza esercitata su  $m_i$  da  $m_j$  è:

$$F_{ij} = m_i m_j \frac{(\underline{r}_j - \underline{r}_i)}{|\underline{r}_j - \underline{r}_i|^3}$$

Le equazioni del moto sono

$$m_i \frac{d^2 \underline{r}_i}{dt^2} = \sum_j m_i m_j \frac{\underline{r}_j - \underline{r}_i}{|\underline{r}_j - \underline{r}_i|^3}$$

con rif. al caso  $i=1$

$$\ddot{\underline{r}}_1 = m_2 \frac{\underline{r}_2 - \underline{r}_1}{d_{12}^3} + m_3 \frac{\underline{r}_3 - \underline{r}_1}{d_{13}^3}$$

$$\left\{ \begin{array}{l} \ddot{x}_1 = m_2 \frac{x_2 - x_1}{d_{12}^3} + m_3 \frac{x_3 - x_1}{d_{13}^3} \\ \ddot{y}_1 = m_2 \frac{y_2 - y_1}{d_{12}^3} + m_3 \frac{y_3 - y_1}{d_{13}^3} \end{array} \right.$$

posto

$$\xi_1 = \dot{x}_1 \quad \text{e} \quad \eta_1 = \dot{y}_1 \quad \text{e' lue:}$$

$$\left\{ \begin{array}{l} \dot{x}_1 = \xi_1 \\ \dot{\xi}_1 = m_2 \frac{x_2 - x_1}{d_{12}^3} + m_3 \frac{x_3 - x_1}{d_{13}^2} \\ \dot{y}_1 = \eta_1 \\ \dot{\eta}_1 = m_2 \frac{y_2 - y_1}{d_{12}^3} + m_3 \frac{y_3 - y_1}{d_{13}^3} \end{array} \right.$$

posto

$$\underline{Y} = [x_1, y_1, \xi_1, \eta_1, x_2, y_2, \xi_2, \eta_2, x_3, y_3, \xi_3, \eta_3]^T$$

e' lue

$$\frac{d}{dt} Y(1,2) = Y(3,4)$$

$$\frac{d}{dt} Y(3,4) = m_2 \frac{Y(5,6) - Y(1,2)}{\|Y(5,6) - Y(1,2)\|^3} + m_3 \frac{Y(9,10) - Y(1,2)}{\|Y(9,10) - Y(1,2)\|^3}$$

$$\frac{d}{dt} Y(5,6) = Y(7,8)$$

$$\frac{d}{dt} Y(7,8) = m_1 \frac{Y(1,2) - Y(5,6)}{\|Y(1,2) - Y(5,6)\|^3} + m_3 \frac{Y(9,10) - Y(5,6)}{\|Y(9,10) - Y(5,6)\|^3}$$

$$\frac{d}{dt} Y(9,10) = Y(11,12)$$

$$\frac{d}{dt} Y(11,12) = m_1 \frac{Y(1,2) - Y(9,10)}{\|Y(1,2) - Y(9,10)\|^3} + m_2 \frac{Y(5,6) - Y(9,10)}{\|Y(5,6) - Y(9,10)\|^3}$$